

SignSee

User Testing Report

June 8, 2026 • 5 Participants • Preliminary Findings

1. Overview

This report summarises the first round of in-person user testing sessions conducted on 8 June 2026 for SignSee, a Dutch Sign Language (NGT) recognition and learning application. Five participants completed live walkthroughs of the app, providing verbal feedback as they explored its features. Two additional survey responses were collected asynchronously. Because the survey is still open, the quantitative results presented here are preliminary and will be updated in the final report.

2. Participants

Five participants took part in moderated screen-share sessions conducted via Microsoft Teams. Session durations ranged from approximately 4 to 10 minutes. Participants are anonymised as P1–P5 in this report. A brief summary of each session is provided below.

Participant	Session Length	Prior Sign Language Knowledge	Notes
P1	~4 min	None	First session of the day; explored all sections independently
P2	~4 min	None	Tried both learning and play sections; positive overall impression
P3	~5 min	None	Tested learning, game, and data collection; noted lighting issues
P4	~10 min	None	Most extended session; engaged deeply with game and data collection
P5	~8 min	None	Tested all sections; expressed strong positive feedback at the end

3. Session Structure

Each session followed a consistent format. The moderator introduced the application as a Dutch Sign Language recognition tool serving as both a learning aid and a gamified practice environment. Participants were then given free rein to explore the app at their own pace, with occasional guidance when navigation required it. The key sections covered were:

- Learn: fingerspelling alphabet with graphical and photographic hand references, real-time camera-based recognition feedback
- Play: random letter game with scoring, hints (textual then visual), and a coin reward system for streaks
- Collect Data: three modes: direct camera capture, image upload, and auto-labelling with automatic folder routing

4. Usability Observations

The following themes emerged consistently across the five sessions.

4.1 Navigation and Layout

- Most participants navigated between sections without assistance once the home page was visible. The purple branded aesthetic and the Otto mascot received unsolicited positive comments from multiple users.
- Back navigation caused confusion for at least two participants (P1, P5). One survey respondent explicitly flagged the absence of a visible back button.
- Greyed-out / unavailable game levels were noticed immediately. P1 asked directly about them; the survey also produced the suggestion to make it clearer why certain modes are unavailable.
- The login screen briefly confused P5, who had to be guided to the “continue without account” option. This is a consequence of testing on a local deployment rather than the live Azure build.

4.2 Learn Section

- Participants found the dual visual reference (illustration + real hand photo) helpful, and several commented positively on it.
- The real-time recognition feedback generated immediate engagement; multiple participants made exclamation sounds or laughed when a sign was recognised correctly.
- Recognition accuracy was variable; lighting conditions in the lab were noted as suboptimal during two sessions. The locally-hosted model is a lighter version of the Azure-deployed one, which was acknowledged to participants.

- Some letters were consistently harder to perform correctly (e.g. N, C, Y). Participants needed repositioning cues such as “face it more towards the camera.”

4.3 Play / Game Section

- The scoring system and coin rewards for correct streaks were understood without explanation by P3 and P4.
- The hint system (textual after ~10 seconds, then visual pop-up) was interacted with by P4. The visual hint pop-up was noted as not yet populated in the current build.
- The 22 supported letters (dynamic/movement-based letters excluded) was explained to P4, who accepted the rationale.
- Participants without prior sign language knowledge found the game motivating but noted that meaningful play requires first completing a learning phase, echoing the intended user journey.

4.4 Collect Data Section

- The three-mode structure (camera, upload, auto-label) was understood after a brief explanation.
- The auto-label flow (sign a letter, wait for model confidence, auto-save within bounding box) was demonstrated successfully with letter A by P4 and P5.
- A usability friction point was noted: pressing a button while holding a hand sign in frame is physically awkward. The 5-second hold-to-save fallback partially addresses this.
- Participants responded positively to the explanation of how collected data feeds a future retraining pipeline.

4.5 General Impressions

- All five participants expressed positive or enthusiastic reactions by the end of their sessions. P5 commented that the interaction and the self-checking aspect were particularly enjoyable.
- No participant reported feeling lost or frustrated with the core flows.
- Several participants played with the app beyond the guided sections, suggesting good intrinsic motivation.

5. Preliminary Survey Results

Three survey responses have been received to date. The survey is still open; this section will be updated in the final report.

Question	R1	R2	R3
Interface clarity (1=very easy, 5=very difficult)	1	5	1
Navigation between pages (1=very easy, 5=very difficult)	2	5	1
Home page experience	Very easy	Moderate	Very easy
Learn page experience	Easy	Moderate	Very easy
Game page experience	Easy	Moderate	Very easy
Data Collection page experience	Not so easy	Moderate	Very easy
Anything confusing or out of place?	Yes: back button to home unclear	No	No
Suggestions for design/layout	Clarify why modes are unavailable	Unsure about target age group	No suggestions; found everything clear and intuitive

With three responses, two out of three rated the interface and navigation at maximum ease, and found all pages very easy to use. The outlier (R2) rated everything as moderate or difficult and was uncertain about the target age group. The back button and unavailable level labelling remain the only flagged usability issues. More responses are needed before drawing firm conclusions, but the early signal is broadly positive.

6. Prioritised Issues

Based on frequency of observation across sessions and survey responses, the following issues are ranked by priority.

Priorit y	Issue	Observed / Reported	Suggested Fix
High	Back/home navigation not obvious	2+ sessions + survey	Add persistent home icon or breadcrumb nav
High	Unavailable game levels lack explanation	2+ sessions + survey	Add tooltip or label (e.g. "Coming soon")
Medium	Login flow confusing in local build	1 session	Surface "continue without account" more prominently
Medium	Holding hand sign while pressing button is awkward	1 session	Extend or surface hold-to-save as default CTA
Medium	Visual hint pop-up not yet populated	1 session	Populate reference images before next test round

Low	Lighting sensitivity affects recognition	2 sessions	Add in-app lighting guidance or indicator
Low	Target audience not clear to new users	1 survey response	Add brief onboarding screen or tagline

7. Positive Signals

Several strengths emerged consistently and should be preserved in future iterations.

- Dual visual references (illustration + photo) in the Learn section were praised and aided sign comprehension.
- Real-time recognition feedback generated immediate delight and kept participants engaged.
- The app's visual identity (purple palette, Otto mascot, overall polish) received unprompted compliments from multiple participants.
- The gamification logic (scoring, coins, hints) was intuitive to users who reached the Play section.
- The data collection concept resonated with participants once the retraining pipeline was briefly explained.

8. Next Steps

- Collect remaining survey responses and update the quantitative analysis.
- Address the high-priority navigation issues (back button, unavailable level labelling) before the next test round.
- Populate visual hint pop-ups for the Game section.
- Consider a short onboarding screen or tooltip tour for first-time users.
- Test the Azure-hosted deployment with users for a fair assessment of model performance.
- Run a second, larger-scale user test round once the above fixes are in place.